CONTINUOUS ASSESSMENT (CA)

Backend Development | Practical Assessment

Total Marks: 8	**Duration:** 1 Hour	**Technology:** Node.js, Express.js, MongoDB

Problem Statement: Student Feedback Management API

An educational institution wants to develop a backend API to collect and manage feedback submitted by students after academic sessions. Each feedback record should contain the student's name, course name, feedback message, rating (1 to 5), and review status. You are required to build a REST API using Node.js, Express.js, and MongoDB with Mongoose to submit, retrieve, and update feedback records.

Subtasks and Marks

Subtask 1: Submit Student Feedback — 3 Marks

Implement **POST /api/feedback** to accept feedback details and save a new record in MongoDB.
• Create a suitable Mongoose schema and model.
• Accept **studentName**, **courseName**, **message**, and **rating** in the request body.
• Validate required fields and ensure rating is between 1 and 5.

Subtask 2: Retrieve Feedback with Pagination — 3 Marks

Implement **GET /api/feedback** to retrieve feedback records from MongoDB.
• Accept **page** and **limit** through query parameters.
• Use **skip()** and **limit()** to implement pagination.
• Return feedback records along with the current page and total number of records.

Subtask 3: Update Feedback Status — 2 Marks

Implement **PATCH /api/feedback/:id** to update the review status of an existing feedback record.
• Accept the new status (for example, **Pending** or **Reviewed**) in the request body.
• Update the record using its MongoDB ID.
• Return an appropriate error response if the feedback record does not exist.

Evaluation Rubric

Criterion	Marks
Mongoose schema/model and feedback submission API	3
Pagination using query parameters, skip() and limit()	2
Pagination metadata and total record count	1
Correct status update API using feedback ID	1
Appropriate HTTP status codes and error handling	1
Total	**8**

Student Instructions

• Use Node.js, Express.js, and MongoDB with Mongoose.

• Test the APIs using Postman, Thunder Client, or another REST client.

• Submit the source code and API response screenshots as instructed.

• Authentication, frontend development, and deployment are not required.

Suggested time plan: Setup and schema — 10 min | Feedback API — 15 min | Pagination — 20 min | Status update and testing — 15 min